

PART - A

JAVA LAB PROGRAMS

1. Write a program to find factorial of list of number reading input as command line arguments.

```
class Factorial
{
    public static void main(String args[])
    {
        int x = args.length; // number of command line arguments
        for (int i = 0; i < x; i++) // loop through each command line argument
        {
            int n = Integer.parseInt(args[i]); // convert the argument to an integer
            if (n < 0)
            {
                System.out.println(n + " is a Negative number.");
            }
            else if (n == 0)
            {
                System.out.println("Factorial of 0 is: 1");
            }
            else
            {
                int fact = 1;
                for (int j = 1; j <= n; j++) // loop to calculate factorial
                {
                    fact = fact * j;
                }
                System.out.println("Factorial of " + n + " is: " + fact);
            }
        }
    }
}
```

Output:

```
javac Factorial.java
java Factorial 5 6 7
Factorial of 5 is: 120
Factorial of 6 is: 720
```

Factorial of 7 is: 5040

```
java Factorial 0
Factorial of 0 is: 1
```

```
java Factorial -3
-3 is a Negative number.
```

2. Write a program to display all prime numbers between two limits.

```
import java.io.*;
class Prime
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        System.out.println("Enter lower limit: ");
        int lower = Integer.parseInt(br.readLine());
        System.out.println("Enter upper limit: ");
        int upper = Integer.parseInt(br.readLine());

        System.out.println("Prime numbers between " + lower + " and " + upper
+ " are: ");
        for (int i = lower; i <= upper; i++)
        {
            int count = 0;
            for (int j = 1; j <= i; j++)
            {
                if (i % j == 0)
                {
                    count++;
                }
            }
            if (count == 2)
            {
                System.out.print(i + " ");
            }
        }
    }
}
```

Output:

```
javac Prime.java
java Prime
Enter lower limit:
10
Enter upper limit:
30
Prime numbers between 10 and 30 are:
11 13 17 19 23 29
```

```
java Prime
Enter lower limit:
1
Enter upper limit:
10
Prime numbers between 1 and 20 are:
2 3 5 7
```

3. Write a program to sort list of elements in ascending and descending order and show the exception handling.

```
import java.io.*;
class Sort
{
    public static void main(String args[]) throws IOException
    {
        try
        {
            int arr[] = new int[10];
            InputStreamReader isr = new InputStreamReader(System.in);
            BufferedReader br = new BufferedReader(isr);

            System.out.println("Enter total number of elements: ");
            int n = Integer.parseInt(br.readLine());
            System.out.println("Enter " + n + " elements one by one: ");
            for (int i = 0; i < n; i++)
```

```

{
    arr[i] = Integer.parseInt(br.readLine());
}
// Add sorting logic here
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        if (arr[i] < arr[j])
        {
            int temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
        }
    }
}

System.out.println("Ascending order: ");
for (int i = 0; i < n; i++)
{
    System.out.print(arr[i] + " ");
}

System.out.println("\nDescending order: ");
for (int i = n - 1; i >= 0; i--)
{
    System.out.print(arr[i] + " ");
}
}
catch (ArrayIndexOutOfBoundsException e)
{
    System.out.println("The number of elements must be less than or
equal to 10 Integers.");
}
}
}

```

Output:

```

javac Sort.java
java Sort
Enter total number of elements:
5
Enter 5 elements one by one:

```

```
20
50
10
40
30
Ascending order:
10 20 30 40 50
Descending order:
50 40 30 20 10
```

4. Write a program to implement all string operations.

```
import java.io.*;
class StrOper
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        System.out.println("***** STRING OPERATIONS *****");
        System.out.println("1. Concatenation of two strings");
        System.out.println("2. Comparison of two strings");
        System.out.println("3. Length of a string");
        System.out.println("4. Converting uppercase to lowercase");
        System.out.println("5. Converting lowercase to uppercase");
        System.out.println("Enter your choice: ");
        int choice = Integer.parseInt(br.readLine());

        String str1, str2, str3;
        int n;
        switch (choice)
        {
            case 1:
                System.out.println("Enter two string: ");
                str1 = br.readLine();
                str2 = br.readLine();
                str3 = str1.concat(str2);
                System.out.println("Concatenation of " + str1 + " and " +
str2 + " is: " + str3);
                break;
            case 2:
```

```

        System.out.println("Enter two string: ");
        str1 = br.readLine();
        str2 = br.readLine();
        n = str1.compareTo(str2);
        if (n == 0)
        {
            System.out.println(str1 + " and " + str2 + " are
equal.");
        }
        else if (n < 0)
        {
            System.out.println(str1 + " is less than " + str2);
        }
        else
        {
            System.out.println(str1 + " is greater than " + str2);
        }
        break;
    case 3:
        System.out.println("Enter a string: ");
        str1 = br.readLine();
        n = str1.length();
        System.out.println("Length of " + str1 + " is: " + n);
        break;
    case 4:
        System.out.println("Enter a string in uppercase: ");
        str1 = br.readLine();
        System.out.println(str1 + " in lowercase is: " +
str1.toLowerCase());
        break;
    case 5:
        System.out.println("Enter a string in lowercase: ");
        str1 = br.readLine();
        System.out.println(str1 + " in uppercase is: " +
str1.toUpperCase());
        break;
    default:
        System.out.println("Invalid choice!");
    }
}
}

```

Output:

```
javac StrOper.java
java StrOper
***** STRING OPERATIONS *****
1. Concatenation of two strings
2. Comparison of two strings
3. Length of a string
5. Converting lowercase to uppercase
Enter your choice:
1
Enter two string:
BCA
Java
Concatenation of BCA and Java is: BCAJava
```

```
java StrOper
***** STRING OPERATIONS *****
1. Concatenation of two strings
2. Comparison of two strings
3. Length of a string
4. Converting uppercase to lowercase
5. Converting lowercase to uppercase
Enter your choice:
3
Enter a string:
computer
Length of computer is: 8
```

5. Write a program to find the area of geometrical figures using method.

```
class Geometrical
{
    public int square(int side)
    {
        return side * side;
    }
    public int rectangle(int length, int breadth)
    {
        return length * breadth;
    }
    public double circle(double radius)
    {
        return 3.14 * radius * radius;
    }
    public double triangle(double base, double height)
    {
        return 0.5 * base * height;
    }
}

class Area
{
    public static void main(String args[])
    {
        Geometrical g = new Geometrical();
        double side = g.square(5);
        double areaOfRectangle = g.rectangle(4, 6);
        double areaOfCircle = g.circle(3.5);
        double areaOfTriangle = g.triangle(4, 5);
        System.out.println("Area of square: " + side);
        System.out.println("Area of rectangle: " + areaOfRectangle);
        System.out.println("Area of circle: " + areaOfCircle);
        System.out.println("Area of triangle: " + areaOfTriangle);
    }
}
```

Output:

```
javac Area.java
java Area
```

Area of square: 25.0
Area of rectangle: 24.0
Area of circle: 38.465
Area of triangle: 10.0

6. Write a program to implement constructor overloading by passing different numbers of parameters of different types.

```
import java.io.*;
class Constructor
{
    String name, regno;
    int mark1, mark2, mark3, total;
    double avg;

    Constructor() // default constructor
    {
        name = "";
        regno = "";
        mark1 = 0;
        mark2 = 0;
        mark3 = 0;
        total = 0;
        avg = 0.0;
    }

    Constructor(String name, String regno, int mark1, int mark2, int mark3)
    // parameterized constructor
    {
        this.name = name;
        this.regno = regno;
        this.mark1 = mark1;
        this.mark2 = mark2;
        this.mark3 = mark3;

        System.out.println("Student Name: " + name);
        System.out.println("Registration Number: " + regno);
        System.out.println("Marks in Subject 1: " + mark1);
        System.out.println("Marks in Subject 2: " + mark2);
        System.out.println("Marks in Subject 3: " + mark3);
    }
}
```

```

    }

    Constructor(Constructor c) // copy constructor
    {
        total = c.mark1 + c.mark2 + c.mark3;
        avg = (double) total / 3.0;
        System.out.println("Total = " + total);
        System.out.println("Average = " + avg);
    }
}

class ExConstructor
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        System.out.print("Enter Student Name: ");
        String name = br.readLine();

        System.out.print("Enter Registration Number: ");
        String regno = br.readLine();

        System.out.println("Enter Marks of three Subjects: ");
        int mark1 = Integer.parseInt(br.readLine());
        int mark2 = Integer.parseInt(br.readLine());
        int mark3 = Integer.parseInt(br.readLine());

        Constructor con = new Constructor(name, regno, mark1, mark2, mark3);
        Constructor con1 = new Constructor(con);
    }
}

```

Output:

```

javac ExConstructor.java
java ExConstructor
Enter Student Name: Ram
Enter Registration Number: 16WXS85058
Enter Marks of three Subjects:
95
98
100

```

Student Name: Ram
Registration Number: 16WXS85058
Marks in Subject 1: 95
Marks in Subject 2: 98
Marks in Subject 3: 100
Total = 293
Average = 97.66666666666667

7. Write a program to create a student report using an applet, read the input using text boxes and display the output using buttons.

StudentReport.java

```
import java.awt.*;
import java.applet.*;
import java.awt.event.*;

public class StudentReport extends Applet implements ActionListener
{
    Label lblRegNo, lblName, lblCourse, lblSem, lblSub1, lblSub2, lblSub3;
    TextField txtRegNo, txtName, txtCourse, txtSem, txtSub1, txtSub2,
    txtSub3;
    Button cmdReport;
    String regNo="", name="", course="", sem="", sub1="", sub2="", sub3="",
    total="", average="";
    int tot;
    float avg;

    public void init()
    {
        // Initialize Labels
        lblRegNo = new Label("Register Number:");
        lblName = new Label("Student Name:");
        lblCourse = new Label("Course:");
        lblSem = new Label("Semester:");
        lblSub1 = new Label("Subject 1:");
        lblSub2 = new Label("Subject 2:");
        lblSub3 = new Label("Subject 3:");

        // Initialize TextFields
```

```

txtRegNo = new TextField(20);
txtName = new TextField(20);
txtCourse = new TextField(20);
txtSem = new TextField(20);
txtSub1 = new TextField(3);
txtSub2 = new TextField(3);
txtSub3 = new TextField(3);

// Initialize Button
cmdReport = new Button("View Report");

// Add components to the Applet
add(lblRegNo);
add(txtRegNo);
add(lblName);
add(txtName);
add(lblCourse);
add(txtCourse);
add(lblSem);
add(txtSem);
add(lblSub1);
add(txtSub1);
add(lblSub2);
add(txtSub2);
add(lblSub3);
add(txtSub3);
add(cmdReport);

// Register action listener
cmdReport.addActionListener(this);
}
public void actionPerformed(ActionEvent e)
{
    if (e.getSource() == cmdReport)
    {
        // Retrieve text input
        regNo = txtRegNo.getText();
        name = txtName.getText();
        course = txtCourse.getText();
        sem = txtSem.getText();
        sub1 = txtSub1.getText();
        sub2 = txtSub2.getText();
        sub3 = txtSub3.getText();

        // Calculate total and average

```

```

        tot = Integer.parseInt(sub1) + Integer.parseInt(sub2) +
Integer.parseInt(sub3);
        avg = (float) tot / 3;

        // Format strings for output display
        regNo = "Register Number: " + regNo;
        name = "Name: " + name;
        course = "Course: " + course;
        sem = "Semester: " + sem;
        sub1 = "Subject 1: " + sub1;
        sub2 = "Subject 2: " + sub2;
        sub3 = "Subject 3: " + sub3;
        total = "Total: " + String.valueOf(tot);
        average = "Average: " + String.valueOf(avg);

        // Redraw applet area to show output
        repaint();
    }
}

public void paint(Graphics g)
{
    // Draw the calculated report below the input form
    g.drawString(regNo, 20, 200);
    g.drawString(name, 20, 220);
    g.drawString(course, 20, 240);
    g.drawString(sem, 20, 260);
    g.drawString(sub1, 20, 280);
    g.drawString(sub2, 20, 300);
    g.drawString(sub3, 20, 320);
    g.drawString(total, 20, 340);
    g.drawString(average, 20, 360);
}
}

```

HTML File

StudentReport.html

```

<html>
<head>
    <title>StudentReport</title>
</head>
<body>

```

```

<center>
    <applet code="StudentReport.class" width="400" height="400">
    </applet>
</center>
</body>
</html>

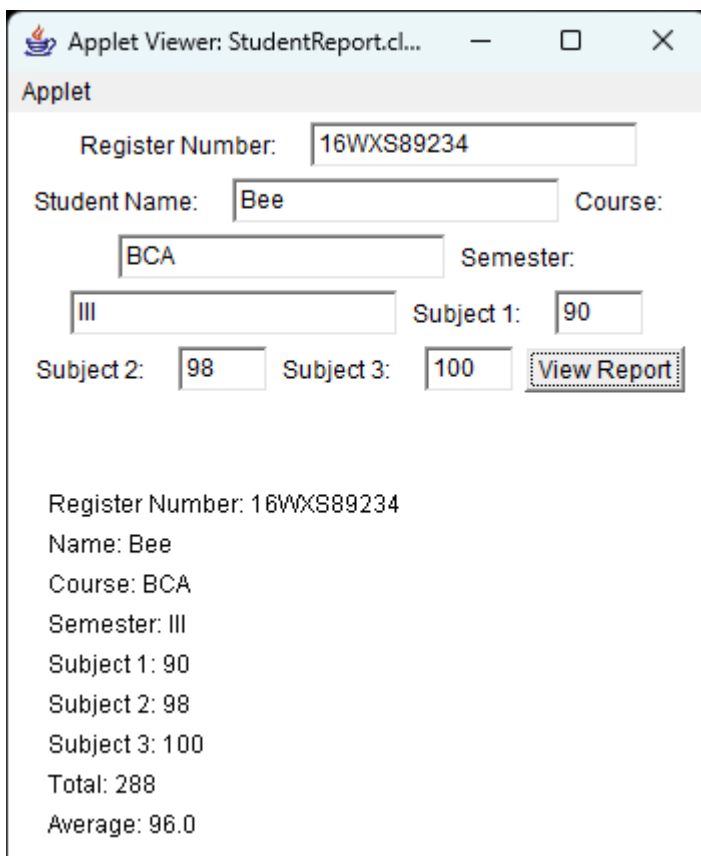
```

Output:

```

javac StudentReport.java
appletviewer StudentReport.html

```



The screenshot shows a Java Applet Viewer window titled "Applet Viewer: StudentReport.cl...". The applet displays a form for entering student information and a section for the generated report.

Form Fields:

- Register Number: 16WXS89234
- Student Name: Bee
- Course: BCA
- Semester: III
- Subject 1: 90
- Subject 2: 98
- Subject 3: 100
- View Report button

Report Output:

```

Register Number: 16WXS89234
Name: Bee
Course: BCA
Semester: III
Subject 1: 90
Subject 2: 98
Subject 3: 100
Total: 288
Average: 96.0

```

8. Write a program to calculate bonuses for different departments using method overloading.

```
abstract class Department
{
    double salary, bonus, netsalary;
    abstract void calBonus(double salary);
    abstract void display();
}

class Accounts extends Department
{
    public void calBonus(double salary)
    {
        this.salary = salary;
        bonus = salary * 0.2;
        netsalary = salary + bonus;
    }
    public void display()
    {
        System.out.println("Accounts \t" + salary + "\t" + bonus + "\t" +
netsalary);
    }
}

class Sales extends Department
{
    public void calBonus(double salary)
    {
        this.salary = salary;
        bonus = salary * 0.3;
        netsalary = salary + bonus;
    }
    public void display()
    {
        System.out.println("Sales \t\t" + salary + "\t" + bonus + "\t" +
netsalary);
    }
}

class Production extends Department
{
    public void calBonus(double salary)
    {
```

```

        this.salary = salary;
        bonus = salary * 0.4;
        netsalary = salary + bonus;
    }
    public void display()
    {
        System.out.println("Production \t" + salary + "\t" + bonus + "\t" +
netsalary);
    }
}

class Bonus
{
    public static void main(String args[])
    {
        double basic[] = {15000.0, 20000.0, 25000.0};
        Department dept[] = new Department[20];
        dept[0] = new Accounts();
        dept[1] = new Sales();
        dept[2] = new Production();

        for (int i = 0; i < basic.length; i++)
        {
            dept[i].calBonus(basic[i]);
        }

        System.out.println("Department \tSalary \tBonus \tNet Salary");
        for (int i = 0; i < basic.length; i++)
        {
            dept[i].display();
        }
    }
}

```

Output:

```

javac Bonus.java
java Bonus

```

Department	Salary	Bonus	Net Salary
Accounts	15000.0	3000.0	18000.0
Sales	20000.0	6000.0	26000.0
Production	25000.0	10000.0	35000.0

9. Write a program to implement thread, applets and graphics by implementing animation of moving ball.

BallMoving.java

```
import java.awt.*;
import java.applet.*;

public class BallMoving extends Applet implements Runnable
{
    Thread T = null;
    int pos = 0;

    public void start()
    {
        T = new Thread(this);
        T.start();
    }

    public void run()
    {
        while (true)
        {
            for (pos = 0; pos < getSize().width; pos = pos + 5)
            {
                repaint();
                try
                {
                    T.sleep(100);
                }
                catch (InterruptedException e)
                {
                }
            }
        }
    }

    public void stop()
    {
        T.stop();
        T = null;
    }

    public void paint(Graphics g)
```

```

{
    g.setColor(Color.blue);
    g.fillOval(pos, 30, 80, 80);

    g.setColor(Color.green);
    g.fillOval(pos + 15, 43, 50, 50);

    g.setColor(Color.red);
    g.drawString("SDC", pos + 25, 65);
    g.drawString("Kolar", pos + 25, 75);
}
}

```

HTML File

BallMoving.html

```

<Html>
<Head>
    <Title>Ball Moving</Title>
</Head>
<Body>
<Center>
    <Applet code=BallMoving.class Width=400 Height=400>
    </Applet>
</Center>
</Body>
</Html>

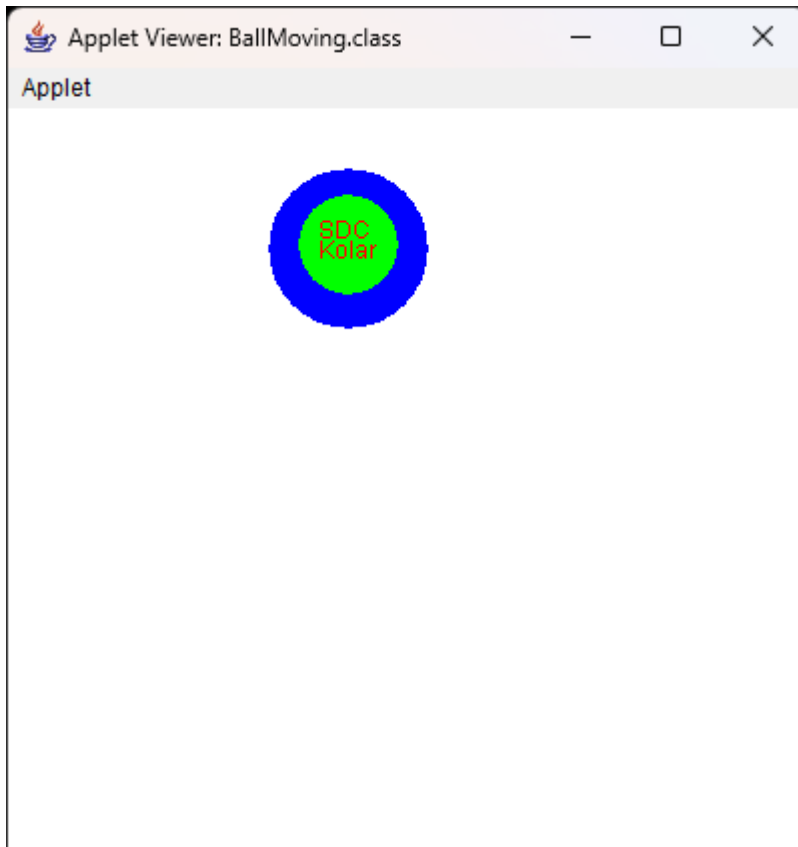
```

Output:

```

javac BallMoving.java
appletviewer BallMoving.html

```



10. Write a program to implement mouse events and keyboard events.

a) Program to implement mouse events using applet.

Mouse.java

```
import java.awt.*;
import java.applet.*;

public class Mouse extends Applet
{
    String msg = "";
    int mx = 0, my = 0;

    public boolean mouseDown(Event obj, int x, int y)
    {
        mx = x;
```

```

        my = y;
        msg = "Mouse Down";
        repaint();
        return true;
    }

    public boolean mouseUp(Event obj, int x, int y)
    {
        mx = x;
        my = y;
        msg = "Mouse Up";
        repaint();
        return true;
    }

    public boolean mouseMove(Event obj, int x, int y)
    {
        showStatus("Moving mouse at " + x + ", " + y);
        return true;
    }

    public boolean mouseDrag(Event obj, int x, int y)
    {
        mx = x;
        my = y;
        msg = "*";
        showStatus("Drawing mouse at " + x + ", " + y);
        repaint();
        return true;
    }

    public boolean mouseEnter(Event obj, int x, int y)
    {
        mx = 10;
        my = 10;
        msg = "Mouse Just Entered";
        repaint();
        return true;
    }

    public boolean mouseExit(Event obj, int x, int y)
    {
        mx = 10;
        my = 10;
        msg = "Mouse Just Left";
    }

```

```
        repaint();
        return true;
    }

    public void paint(Graphics g)
    {
        g.drawString(msg, mx, my);
    }
}
```

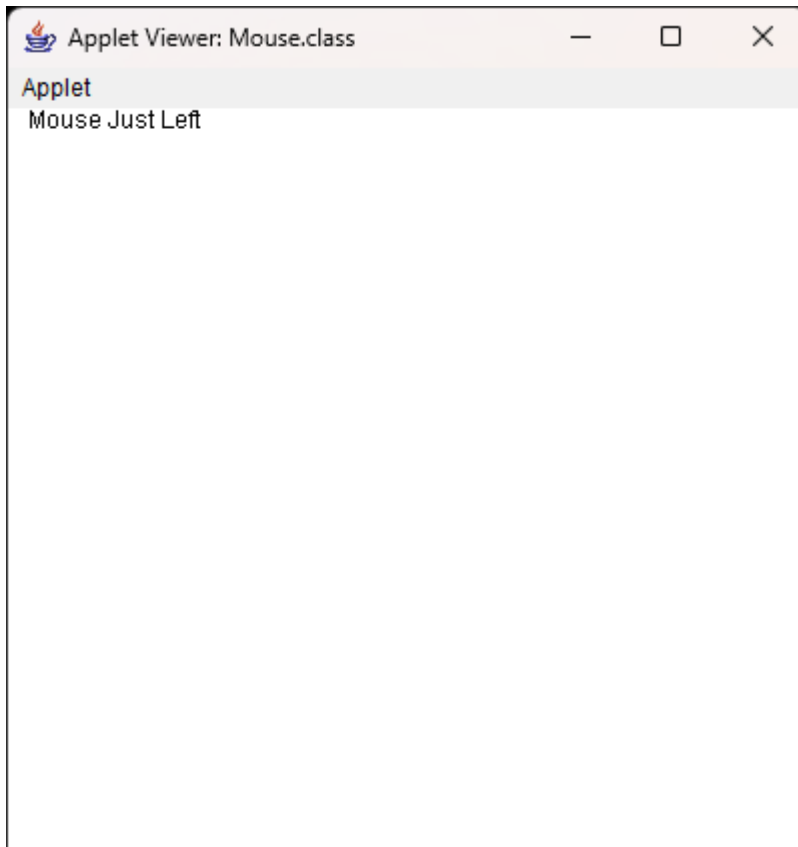
HTML File

Mouse.html

```
<Html>
<Head>
    <Title>Mouse Events</Title>
</Head>
<Body>
<Center>
    <Applet code=Mouse.class Width=400 Height=400>
    </Applet>
</Center>
</Body>
</Html>
```

Output:

```
javac Mouse.java
appletviewer Mouse.html
```



b) Program to implement keyboard events using applet.

Keyboard.java

```
import java.awt.*;
import java.applet.*;

public class Keyboard extends Applet
{
    String msg = "";

    public boolean keyDown(Event obj, int key)
    {
        msg = msg + (char)key;
        repaint();
        showStatus("Key Down");
        return true;
    }

    public boolean keyUp(Event obj, int key)
    {
        showStatus("Key Up");
    }
}
```

```

        return true;
    }

    public void paint(Graphics g)
    {
        g.drawString(msg, 10, 20);
    }
}

```

HTML File

Keyboard.html

```

<Html>
<Head>
    <Title>Keyboard Events</Title>
</Head>
<Body>
<Center>
    <Applet code=Keyboard.class Width=400 Height=400>
    </Applet>
</Center>
</Body>
</Html>

```

Output:

```

javac Keyboard.java
appletviewer Keyboard.html

```

